PAPER_1605 — DNA Base Pairs per Helical Turn = bp/turn = $SO_5 + F \cdot D + F^2 \cdot SO_5 = 10 + 0.4 + 0.1 = 10.5$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Biology **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

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Observation

PAPER_1209BB_S601 (Biology Unified Proof Set) lists **DNA Base Pairs per Helical Turn** with the integer-primitive expression:

```
bp/turn = SO_5 + F·D + F²·SO_5 = 10 + 0.4 + 0.1 = 10.5 EXACT
```

Residual: **EXACT**.

UQFF Closed Identity

```
bp/turn = SO_5 + F·D + F²·SO_5 = 10 + 0.4 + 0.1 = 10.5 EXACT    EXACT
```

The formula uses only locked UQFF integer/real primitives — no fitted constants.

DNA Structure — 10.5 bp per Helical Turn EXACT

The Watson-Crick B-DNA double helix has exactly 10.5 base pairs per full turn (3.4 Å rise per bp × 34 Å pitch). This is fundamental to DNA's ability to wrap around histones, form chromatin, and execute genetic transcription.

UQFF's structural prediction:

```
bp/turn = SO_5 + F_TRZ·D_phys + F_TRZ²·SO_5
        = 10 + 0.4 + 0.1
        = 10.5 EXACT
```

The fundamental geometric constant of life's molecular machinery sits at SO_5 + small corrections. UQFF's integer primitives govern both **cosmological recombination redshift (z=1090)** and **DNA helical pitch (10.5 bp/turn)** — spanning ~30 orders of magnitude in physical scale from megaparsec to nanometer.

NOT REPLACEMENT

Empirical biology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209BB_S601
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "dna_bp_per_turn_10_5"})`
